=== Famicom bus interface (FC-EXPSYNTH additions) ===

J10: console-side gold fingers (1-30 front / 31-60 back)

J11: 60-pin slot for the expansion cartridge (pin n -> pin n: /ROMSEL gated)

Primary Flash

Secondary Flash/PSRAM

Optional Secondary flash or PSRAM (U4), using the second chip select (XIP_SS_N[1]) on GPIO0.
N.B. Pull-up R13 will be required as GPIOs default to pull-downs at power-up.

12MHz Crystal

Raspberry Pi RP2350B QFN-80

Make sure R7 and R8 are close to RP2350

JST Debug Connector

U10-U12: 5V bus -> 3.3V (74LVC245, VCC=3V3, DIR=GND: B->A, /OE=GND)

U13: D0-D7 3.3V->5V (74AHCT245, DIR=5V: A->B)

U14/U15: /ROMSEL gating and U13 output enable

ROMSEL_EXP = ROMSEL_N OR R/W (J11 pin44)
RDSEL_N = ROMSEL_N OR NOT R/W
EN_N = RDSEL_N OR EN_N (U13 /OE)

EN_N: GPIO27 low = U13 enabled. 10k pull-up keeps it disabled at reset.

Power: console +5V and USB VBUS -> Schottky OR -> VSYS -> 3.3V LDO (IOVDD is up whenever console 5V is present)

D10: console +5V -> VSYS

D11: VBUS -> VSYS (blocks back-feed)

+5V sense (GPIO40/ADC0)

Status LED, microSD (SPI0: GPIO32-35, CD=GPIO36), spare header

J14: 6 spare GPIO (GPIO30/31/43-47 left unrouted to ease the QFN fan-out); GPIO0 is left free (PSRAM CS option via R11)

Takara (RP2350B core: Raspberry Pi Ltd minimal design REV2)
Sheet: /
File: FC-EXPSYNTH.kicad_sch
Title: FC-EXPSYNTH: Famicom expansion-audio relay cartridge (RP2350B)
Size: A1 | Date: 2026-11-03 | Rev: E0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1